

Single choice Question with one option correct

Q.01 to Q.10

Multiple choice Question with one or more than one may be correct

Q.11 to Q.30

Numerical Based Question (no negative marking)

Q.31 to Q. 50

Stem Question

Q.51 to Q. 54

DPP Syllabus : IUPAC Nomenclature, POC, GOC-1, 2, 3, Structural & stereoisomerism

1. Two isomers of butyne can be distinguished by

- (A) $\text{H}_2/\text{Pd BaSO}_4$ (B) $\text{Br}_2 / \text{H}_2\text{O}$ (C) $\text{Cu}_2\text{Cl}_2/\text{NH}_4\text{OH}$ (D) Bayer's reagent

2. How many possible isomeric alkenes (including stereoisomers) give 2, 3-Dimethylpentane on catalytic hydrogenation ?

- (A) 4 (B) 6 (C) 8 (D) 5

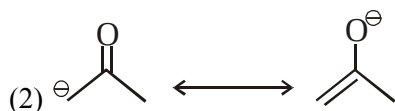
3. Column - I

Column - II

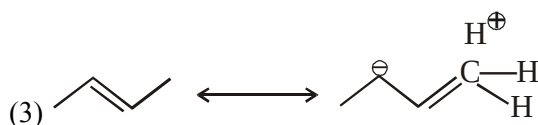
(Orbital overlaps)



(P) p(filled) - π^* (vacant)

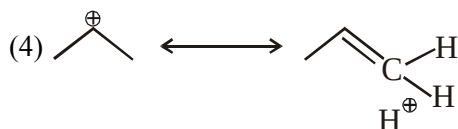


(Q) σ (filled) - p(vacant)



(R) π (filled) - p(vacant)

(S) p(filled) - π (vacant)



(T) σ (filled) - π^* (vacant)

(A) (1)→R ; (2)→P ; (3)→T ; (4)→Q

(B) (1)→P ; (2)→R ; (3)→Q ; (4)→T

(C) (1)→R ; (2)→Q ; (3)→T ; (4)→P

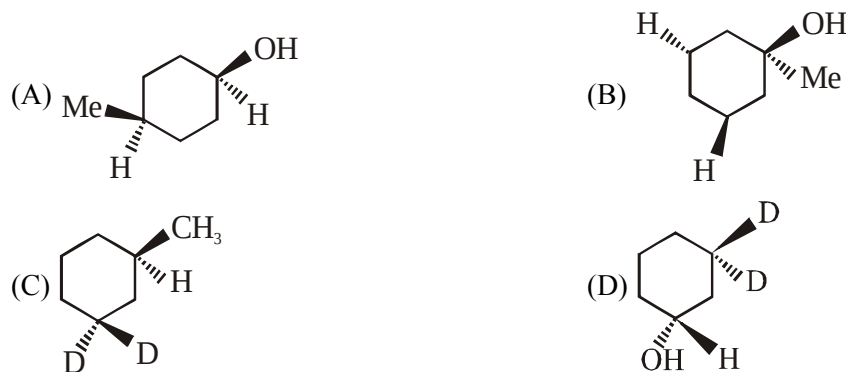
(D) (1)→S ; (2)→R ; (3)→P ; (4)→Q

4.	List-I (Compound)	List-II (Statements)
(P)		(1) It can show tautomerism, optical isomerism but not geometrical isomerism.
(Q)		(2) It can show tautomerism, geometrical isomerism but not optical isomerism.
(R)		(3) It can show geometrical isomerism, optical isomerism but not tautomerism.
(S)		(4) It can show tautomerism, geometrical isomerism & also optical isomerism.

Code :

	P	Q	R	S
(A)	2	1	3	4
(B)	2	3	4	1
(C)	3	2	4	1
(D)	2	3	1	4

5. Racemic mixture of tartaric acid can be resolved by addition of -



6. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.

I. $\text{Me}-\text{CH}=\text{C}=\text{CH}-\text{Br}$ is optically active.

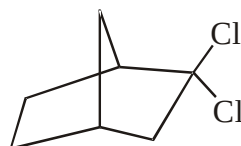
II. All optically active compounds are chiral.

III. All chiral compounds are optically active.

IV. $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{COOH}$ and $\text{CH}_3-\underset{\text{COOH}}{\text{CH}}-\text{CH}_3$ are positional isomers.

(A) FTFF (B) FTFT (C) TFTF (D) FTTF

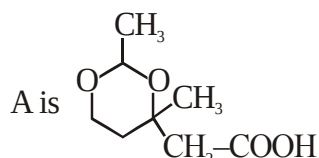
7. Sum of total number of chiral centres (x) and total number of pair of enantiomers (y) for following structure is :-



(A) 2 (B) 3 (C) 4 (D) 5

Comprehension

A research scholar synthesised the compound A in the laboratory.



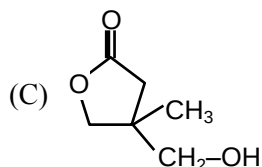
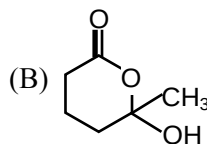
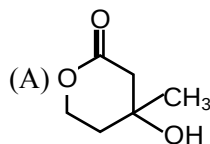
He wanted to study this compound in acidic medium and kept some time for heating. After an hour, he surprised because he got the mixture of two different compounds B and C. He studied their properties and compared them with A.

	A	B	C
Reaction with NaHCO_3	$\text{CO}_2 \uparrow$	No reaction	No reaction
Blue litmus	red	No	No
NaOI	No	No	Yellow ppt
$[\text{Ag}(\text{NH}_3)_2]^+$	No	No	Silver mirror

[H int : $\text{A} \xrightarrow{\text{H}_3\text{O}^+/\Delta} \text{B} + \text{C}$]

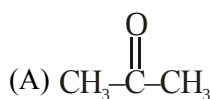
On the basis of above information answer the following questions (8, 9 and 10).

8. Compound B is :-

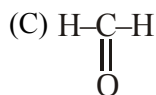


(D) None

9. Compound C is :-



(B) $\text{CH}_3-\text{CH}_2-\text{CH}=\text{O}$



(D) $\text{CH}_3-\text{CH}=\text{O}$

10. B is formed due to :-

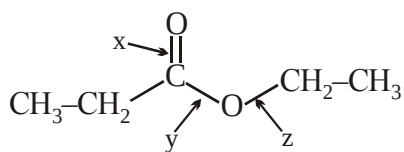
(A) Cleavage of C–O bond followed by esterification between $-\text{COOH}$ & $-\text{OH}$ groups.

(B) Decarboxylation of $-\text{COOH}$ group

(C) Reduction of $-\text{CH}_2\text{COOH}$ into $-\text{CH}_2\text{CH}_2\text{OH}$

(D) None

11. Correctly matched option(s) for following compound is/are :



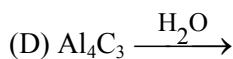
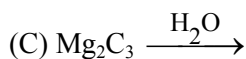
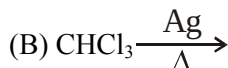
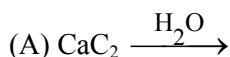
(A) $x > y > z$ (Bond order)

(B) Ethyl ethanoate (IUPAC name)

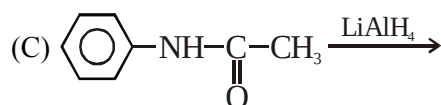
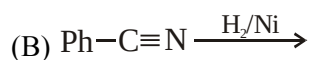
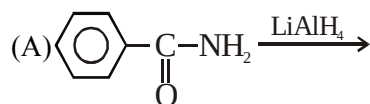
(C) $x > y > z$ (Bond energy)

(D) $z > y > x$ (Bond length)

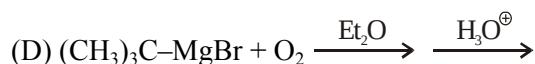
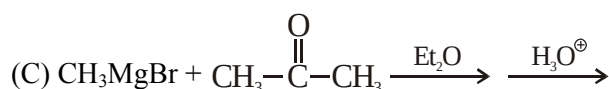
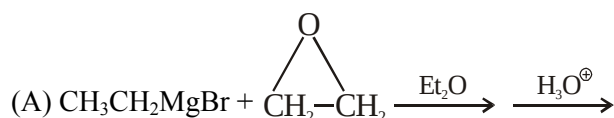
12. Which of the following reaction gives a product, which gives white ppt. with $\text{AgNO}_3 / \text{NH}_4\text{OH}$?



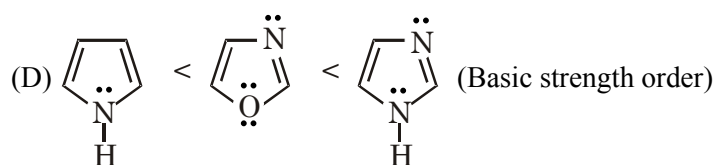
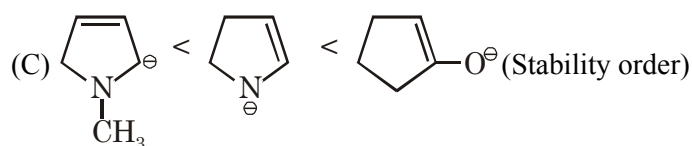
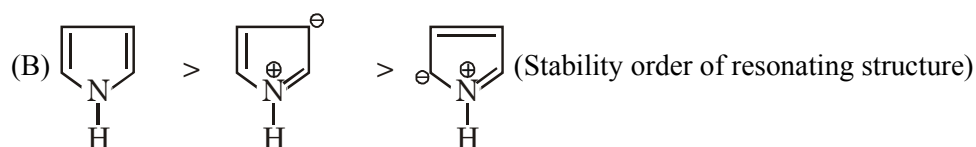
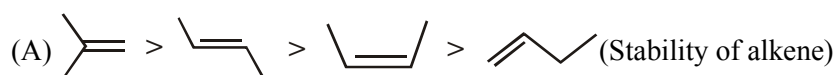
13. Find out the reaction in which obtained product will give positive isocyanide test :



14. Which of the following reaction sequences would be the best for synthesis of t-butyl alcohol?



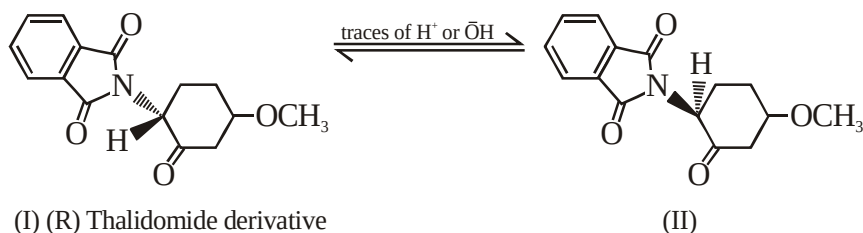
15. Select the order which is correct among following :



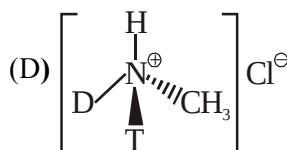
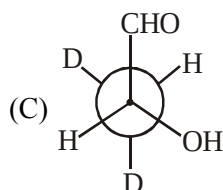
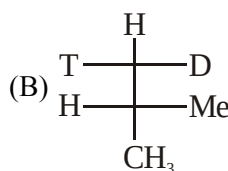
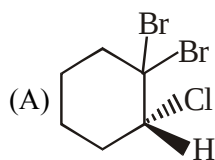
16. Correct statement is/are regarding carboxylic acids having molecular formula ($\text{C}_8\text{H}_8\text{O}_2$) containing benzenoid structure.

- (A) The total number of carboxylic acids (with one benzenoid ring) is 4.
 (B) Out of the three Toluic acids, ortho isomer is the strongest acid.
 (C) All Toluic acids are weaker acids than benzoic acid.
 (D) All Toluic acids are weaker acids than H_2CO_3 .

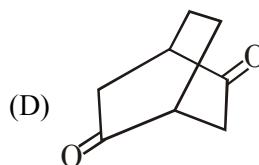
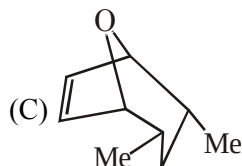
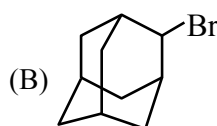
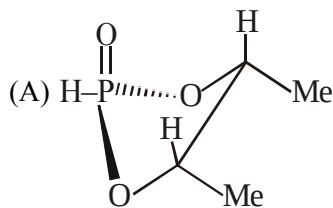
17. Select the correct statement(s) about the following reaction :-



- (A) It is an isomerisation reaction.
- (B) (I) and (II) are diastereomers.
- (C) The reaction is passing through the process of enolisation.
- (D) I and II are tautomers.
18. Which of the following compounds will show geometrical isomerism ?
- (A) $\text{D}-\text{C}\equiv\text{C}-\text{C}=\text{C}-\text{D}$
- (B) $\text{O}=\text{S}$ (cyclic structure)
- (C)
- (D) 1-Chloropent-1-ene
19. Which enamine of molecular formula $\text{C}_5\text{H}_{11}\text{N}$ can exhibit geometrical isomerism ?
- (A)
- (B)
- (C)
- (D)
20. Which of the following compound(s) have R configuration?



21. Identify compounds having plane of symmetry?



22. Which is/are correct combination for the names of given compounds in column-I ?

Column-I (Compounds)	Column-II (Common name)	Column-III (IUPAC name)
(1) $(\text{CH}_3)_2\text{C} = \text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$	(p) Phorone	(w) 2,6-Dimethylhepta-2,5-dien-4-one
(2) $(\text{CH}_3)_2\text{C} = \text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}=\text{C}(\text{CH}_3)_2$	(q) Mesityl oxide	(x) 4-Methylpent-3-en-2-one
(3)	(r) Salol	(y) Phenyl 2-hydroxybenzoate
(4)	(s) Aspirin	(z) 2-Ethanoyloxybenzoic acid

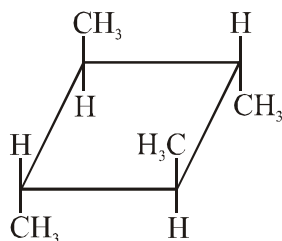
(A) 1, q, x

(B) 2, r, w

(C) 3, s, z

(D) 4, p, y

23. Types of symmetry present in the following molecule is/are :-



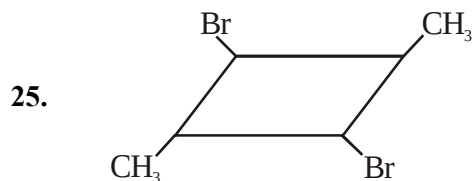
(A) Alternating axis of symmetry

(B) Plane of symmetry

(C) Axis of symmetry

(D) Centre of symmetry

24. Mark the compound given below which have at least one meso stereoisomers :-
- (A) Cyclopentane-1, 3-dicarbaldehyde (B) Hexane-1,2,3,4,5,6-hexaol
(C) 2,3,4,5,6-pentahydroxyhexanal (D) 2,3-Dihydroxybutanedioic acid

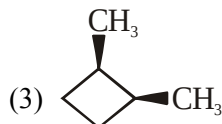
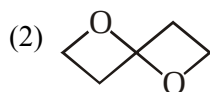
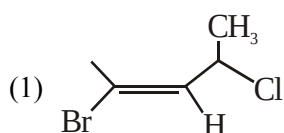


Choose the correct option/s for the given compound.

- (A) Total stereoisomers are five.
(B) All stereoisomers are chiral.
(C) All stereoisomers are achiral.
(D) Four stereoisomers have plane of symmetry & one has centre of symmetry.
26. Match the column (I) and (II) :-

Column-I

Molecule



Column-II

Property

(p) Chiral atom is present

(q) Molecule is chiral

(r) Plane of symmetry is present

(s) Centre of symmetry is present

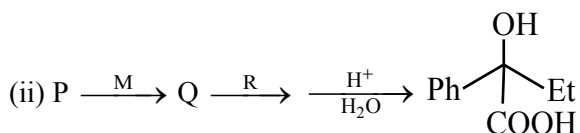
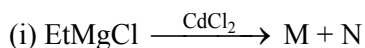
(A) (1) → p ; (2) → q ; (3) → p ; (4) → r

(B) (1) → q ; (2) → q ; (3) → r ; (4) → s

(C) (1) → r ; (2) → q ; (3) → p ; (4) → s

(D) (1) → q ; (2) → p ; (3) → r ; (4) → s

27. Observe the following reactions :



Correct reagents and products is/are :

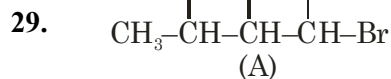
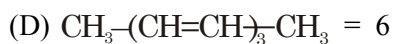
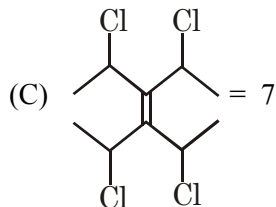
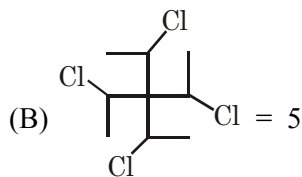
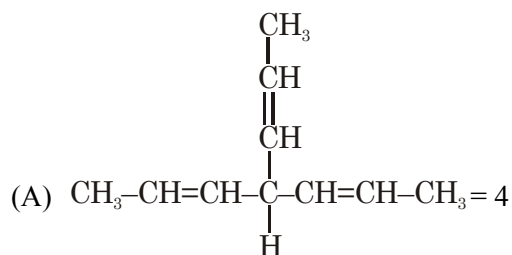
(A) $\text{M} = \text{C}_2\text{H}_5\text{Cd}$

(B) $\text{P} = \text{C}_2\text{H}_5\text{COCl}$

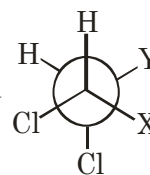
(C) $\text{Q} = \text{PhCOC}_2\text{H}_5$

(D) $\text{R} = \text{KCN}$ or NaCN

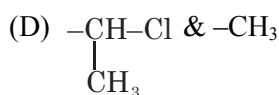
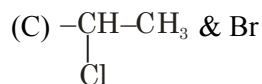
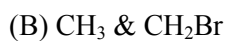
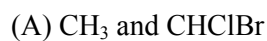
28. Identify compounds correctly matched with their number of stereoisomers :



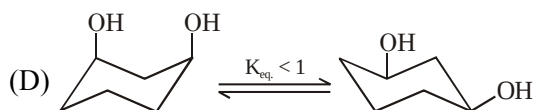
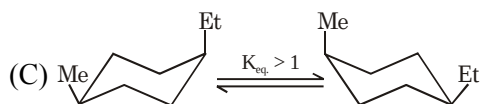
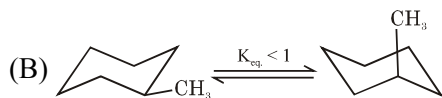
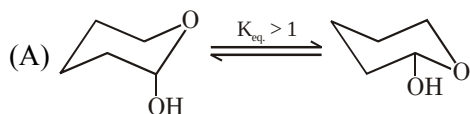
For the above compound (A) the given newmann projection can be represented



X & Y are respectively :



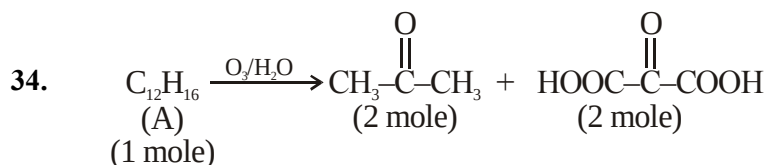
30. Identify compounds or reactions correctly matched with K_{eq} ?



31. What is the number of all the structural isomers of octane with five carbons in their principal chain ?
32. Number of structural isomers possible from the molecular formula $C_2H_2FClBrI$ is :
33. Total number of structural isomers of ether possible for molecular formula $C_5H_{12}O = \textcircled{N_1}$:

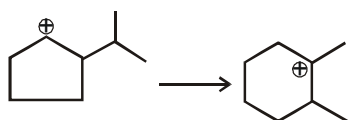
Total number of structural isomers possible for molecular formula $C_4H_9Cl = \textcircled{N_2}$:

value of $(N_1 + N_2)$ is

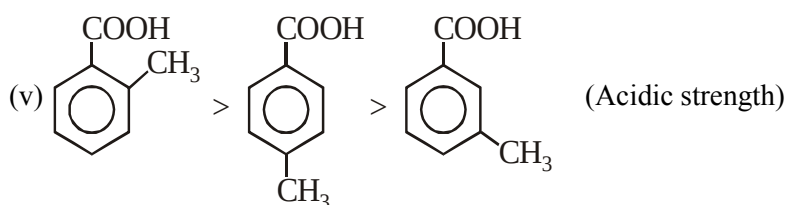
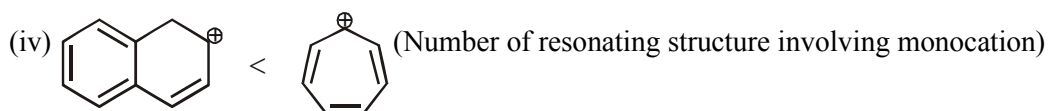
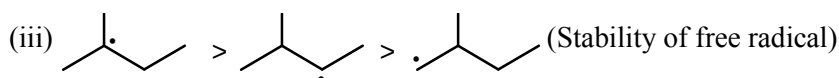
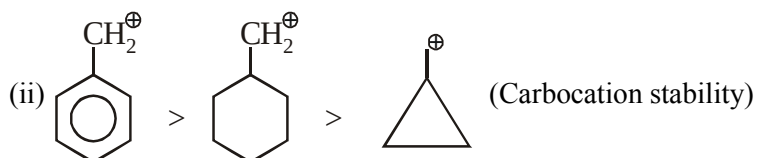
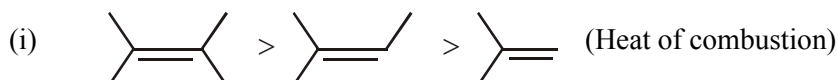


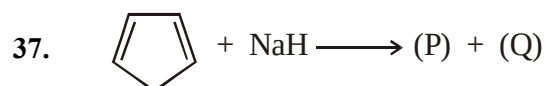
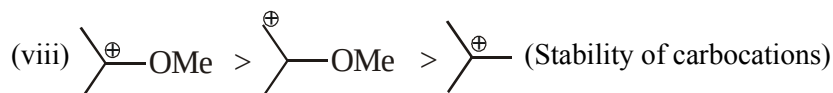
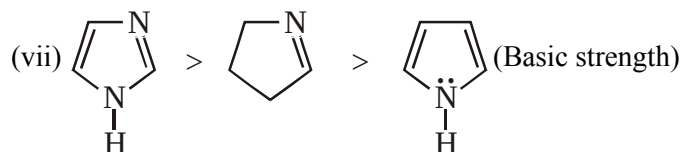
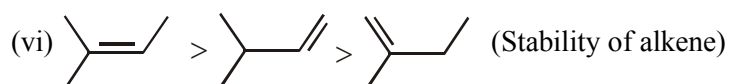
How many π bonds are present in compound A ?

35. How many 1,2-shifts are involved in the following rearrangement of carbocation ?



36. How many of the following orders are correctly matched?





If X = Number of resonating structure of (P) involving monoanion &

Y = Molecular weight of gas (Q).

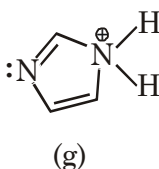
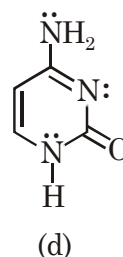
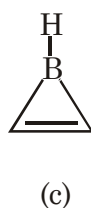
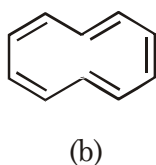
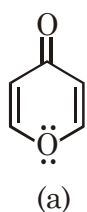
Then find out the value of (X + Y) :

38. If Number of aromatic compounds are X.

Number of non aromatic compounds are Y.

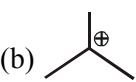
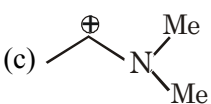
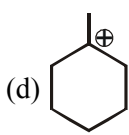
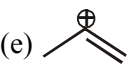
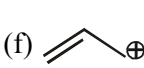
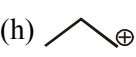
and number of anti aromatic compounds are Z.

Then value of (X + 2Y - Z) is :



39. How many benzenoid structural isomers of C_9H_8O are possible, which give white precipitate with tollens' reagent ?

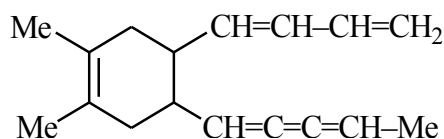
40. How many of the following carbocation is more stable than $\text{CH}_3-\overset{\oplus}{\text{C}}\text{H}-\text{CH}_3$?

- (a) $\text{CH}_3-\text{CH}_2^\oplus$ (b)  (c) 
 (d)  (e)  (f) 
 (g) $\text{Ph}-\overset{\oplus}{\text{C}}\text{H}_2$ (h) 

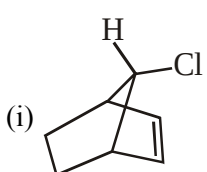
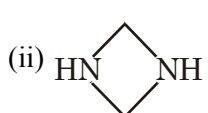
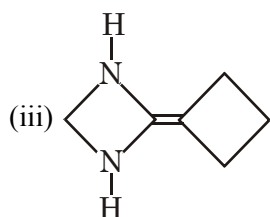
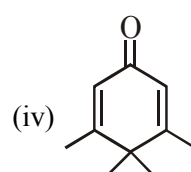
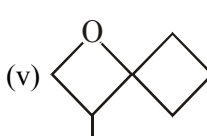
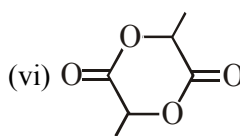
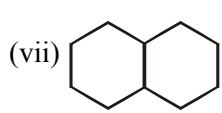
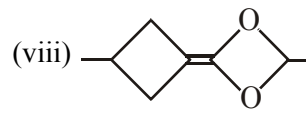
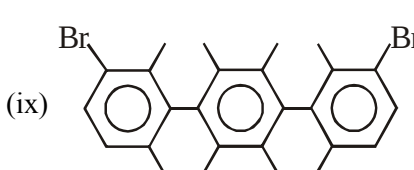
41. How many of the following are less basic than $\text{Et}-\text{NH}_2$?

- (i) NH_2-Cl (ii) $\text{CH}_3-\text{CH}_2-\text{OH}$ (iii) $\text{CH}_3-\text{CH}_2-\text{NH}^\ominus$ (iv) $\text{O}_2\text{N}-\text{CH}_2-\text{NH}_2$
 (v) $\text{F}_3\text{C}-\text{CH}_2-\text{NH}_2$ (vi) $\text{Et}-\text{NH}-\text{CH}_3$ (vii) $\text{Et}-\text{NH}-\text{Et}$ (viii) CH_3-NH_2

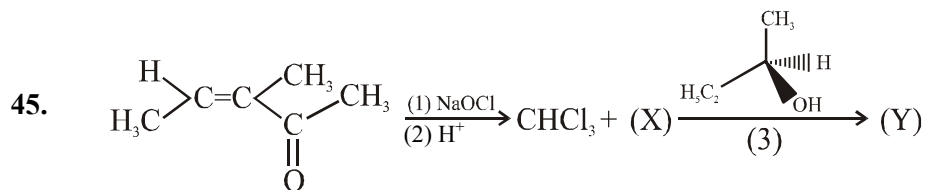
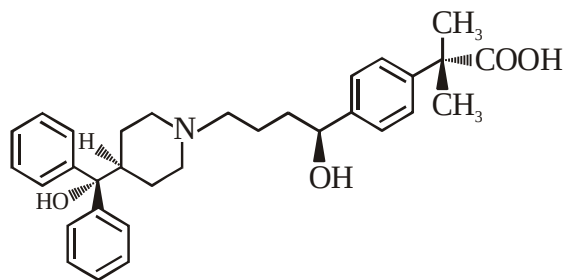
42. How many stereoisomers are possible for the following?



43. How many compounds are there which can show geometrical isomerism ?

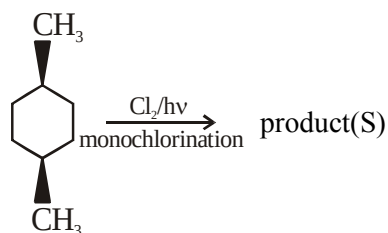
- (i)  (ii)  (iii)  (iv) 
 (v)  (vi)  (vii)  (viii) 
 (ix) 

44. Allegra, a common prescription drug with the structure shown below, is given for the treatment of seasonal allergies. How many chiral carbon does Allegra possess?



The total number of isomeric organic products (Y) formed is : -

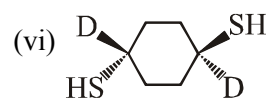
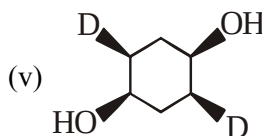
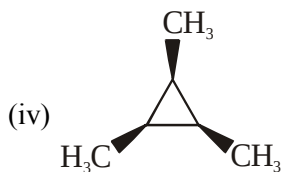
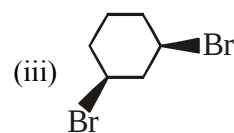
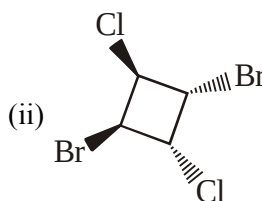
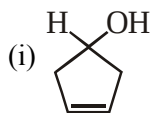
46. How many optically active compounds are possible in the following reaction ?

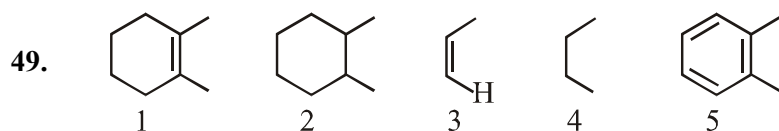


47. A smallest, optically active, acyclic alkane can be obtained by catalytic hydrogenation of any of the five structural isomers of an alkene. How many chiral alkenes gives the above alkane?
48. Calculate sum of X and Y in the given molecules.

X = Number of molecules having plane of symmetry

Y = Number of molecules having centre of symmetry



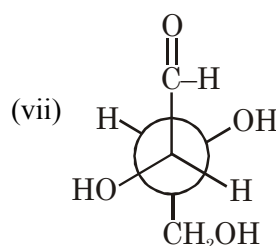
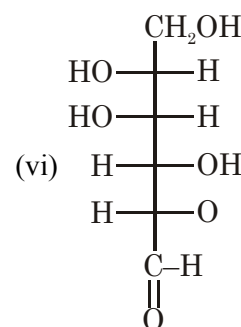
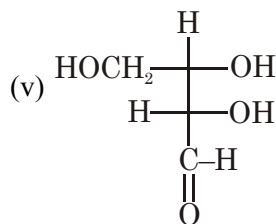
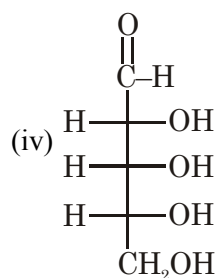
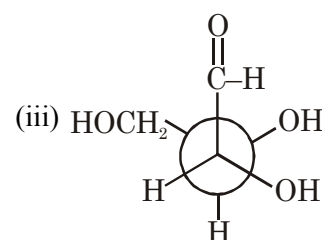
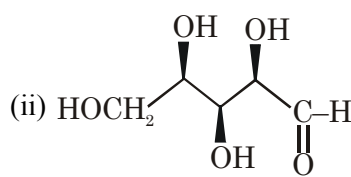
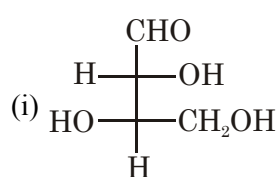


X : Number of that compound which show geometrical isomerism.

Y : Number of that compound which show optical isomerism.

Write the sum of $X + Y = ?$

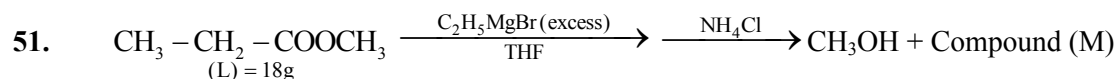
50. Identify total number of following compounds having "D" configuration :



Stem-1

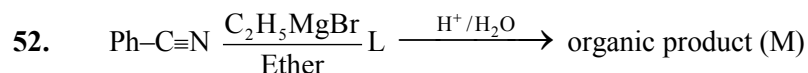
Grignard reagent is an organo metallic compound and a very versatile reagent in the synthesis of organic reactions.

Grignard reagent has carbanionic character and behave like strong nucleophile and strong base. Grignard reagent gives addition reaction with carbonyls, O_2 , CO_2 , alkanenitriles (RCN) etc. while substitution reaction with acid derivatives, alkyl halides, epoxide etc.



Compound L is taken 18g then find the mass obtained of product (M) if percentage yield of final product is 90%.

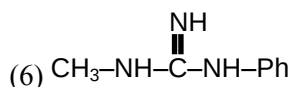
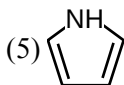
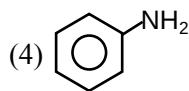
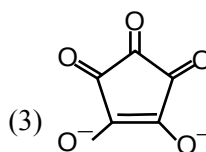
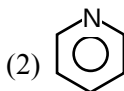
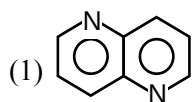
[Consider atomic mass of C=12, H=1, O=16, Br=80. Mg=24]



Molecular mass of the product M is-

Stem-2

Some species are given below to study their characteristics and properties.



53. Select the species amongst above whose all valid resonance structures are identical. Find the total number of resonance structures of that species.
54. Find the molecular mass of the most basic species amongst the given species in stem.

Answer key

1.	(C)	2.	(C)	3.	(A)	4.	(B)	5.	(D)
6.	(A)	7.	(B)	8.	(A)	9.	(D)	10.	(A)
11.	(A,C,D)	12.	(A,B,C)	13.	(A,B)	14.	(B,C,D)	15.	(A,C,D)
16.	(A,B)	17.	(A,B,C)	18.	(A,B,D)	19.	(A,B,C)	20.	(A,B,C)
21.	(A,B,C)	22.	(A,C)	23.	(A,B,C)	24.	(A,B,D)	25.	(A,C,D)
26.	(A,B)	27.	(C,D)	28.	(A,B,C,D)	29.	(A,C)	30.	(A,B,C,D)
31.	6	32.	7	33.	10	34.	4	35.	4
36.	3	37.	7	38.	7	39.	19	40.	5
41.	5	42.	16	43.	7	44.	1	45.	1
46.	4	47.	2	48.	6	49.	4	50.	6
51.	20.88	52.	134.0	53.	5	54.	149		